

IDS Pulse — Product Designer Specification & Design System Guide

A comprehensive visual language, user persona architecture, and engineering guardrail blueprint for product designers and UI/UX specialists building the IDS Pulse platform.

Organization: Integrity Driven Solutions Inc. (IDS) **Target:** Desktop Web + Android Native PWA

Production URL: <https://proud-lavoisier.vercel.app> **Status:** Active Standard

01 EXECUTIVE OVERVIEW & CORE MISSION

Mission: IDS Pulse is the mission-critical digital operating platform for Integrity Driven Solutions Inc. (IDS), powering real-time quality containment, defect inspection, and liaison engineering across major Tier-1 automotive assembly plants (GM, Ford, Stellantis, Tesla, Magna) in the US and Canada.

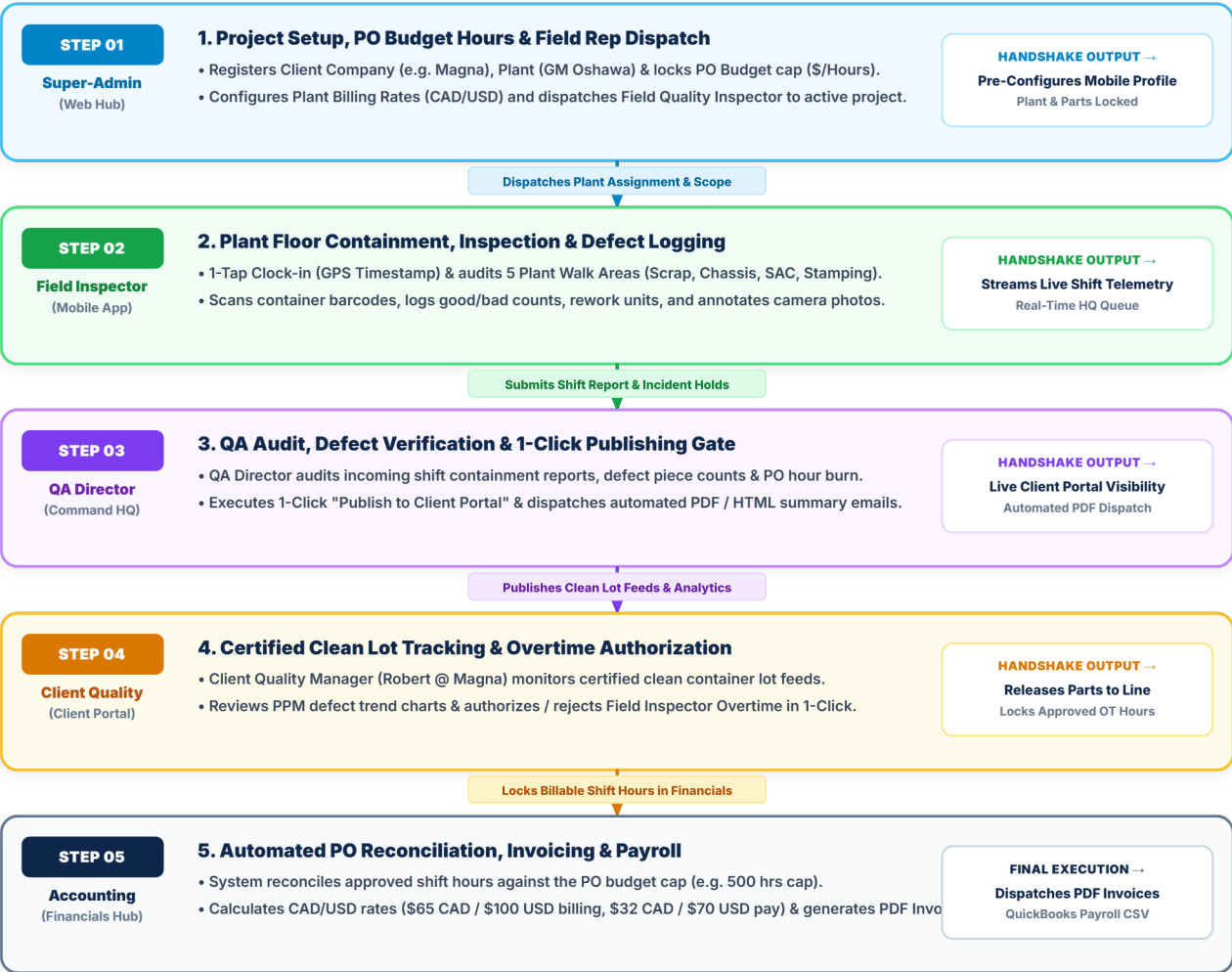
In high-volume automotive manufacturing, a defect containment delay of even 15 minutes can halt an assembly line costing upwards of \$50,000 per minute. The design of IDS Pulse must prioritize **immediate legibility, high contrast, zero ambiguity, and frictionless speed.**

IDS PULSE PLATFORM ECOSYSTEM: CONNECTED HUB & SPOKE NETWORK BLUEPRINT



END-TO-END OPERATIONAL LIFECYCLE: 5-STEP MASTER PROCESS FLOWCHART

Sequential data and operational handshake across all 4 User Roles from initial Project Onboarding to Final Invoicing & Payroll.



02 THE 4 CORE USER PERSONAS

Every feature and UI element must be designed with the exact mental model of the target role:

1. FIELD QUALITY INSPECTOR

MOBILE APP

STAFF DISPATCHED ON PLANT FLOORS

- **Environment:** Loud plant floor, safety gloves, dirty screens, one-handed operation.
- **Key Tasks:** 1-Tap Clock-in, count good/bad parts, photo markup, defect logs.
- **Design Needs:** Touch targets $\geq 44\text{px}$, large fonts, offline queue resilience, zero nested menus.

2. CLIENT QUALITY MANAGER

EXECUTIVE PORTAL

TIER-1 SUPPLIER / OEM CONTACTS

- **Environment:** Office/laptop or tablet on plant walk-throughs.
- **Key Tasks:** Track containment lots, inspect certified clean parts, approve overtime.
- **Design Needs:** High-level visual KPI feeds, instant overtime action drawer, zero internal cost data.

3. QA DIRECTOR / SUPER-ADMIN

COMMAND CENTER

DONNA / GREG / SHAHROZ

- **Environment:** Dual-monitor HQ operations desk.
- **Key Tasks:** Onboard clients/plants, provision logins, dispatch reps, publish reports.
- **Design Needs:** 1-Stop Operations Hub with Master Matrix table, 1-click publishing gates.

4. FINANCE & BILLING CONTROLLER

ACCOUNTING HUB

BACK-OFFICE FINANCIAL AUDITOR

- **Environment:** Accounting spreadsheets & billing reconciliations.
- **Key Tasks:** Timesheet verification, PO cap tracking, batch invoice generation.
- **Design Needs:** Tabular financial drilldowns, CAD vs USD currency tags, PO cap alert bars.

03 DESIGN TOKENS & COLOR SYSTEM

All UI elements enforce strict high-contrast light theme standards with contrast ratios $\geq 7:1$ (WCAG AAA compliant).

Industrial Navy
#10284A (Headers/Nav)

Action Cobalt
#1769E0 (Buttons/Links)

Quality Pass
#059669 (Approved/OK)

Critical Defect
#DC2626 (Scrap/Alert)

TYPOGRAPHY SCALE

ROLE	CSS SIZE & WEIGHT	PIXEL EQUIV	USAGE
Display Title (H1)	text-2xl font-black	24px / Bold 900	Page headers, primary screen titles
Section Title (H2)	text-lg font-black uppercase	18px / Bold 900	Major card & table section headers
Sub-Header (H3)	text-[14px] font-extrabold uppercase	14px / Bold 800	Modal titles, widget header bars
Body Text	text-[13px] font-medium	13px / Medium 500	Standard descriptions, narratives, table cells
Badges & Pills	text-[10px] font-black uppercase	10px / Bold 900	Status tags, currency pills, role badges

04 MANDATORY ENTERPRISE DESIGN GUARDRAILS

Every mock-up, layout, and component MUST obey these non-negotiable rules:

RULE 1

Zero Emoji Policy (Permanent)
Emojis are strictly forbidden across all buttons, feeds, modals, reports, and badges. Use clean SVG icons from lucide-react or crisp typography.

RULE 2

Zero Dark Container Leaks (High-Contrast Light Theme)
Dark container classes (bg-slate-950, bg-amber-950) inside light surfaces are banned. All form inputs must render pure white backgrounds (bg-white border-2 border-slate-300) with dark text.

RULE 3 Zero Text Truncation for Quality Data

Ellipsis (...) is prohibited for defect narratives, containment actions, and shift summaries. Layouts must wrap text dynamically into multi-line rows.

RULE 4 Strict Separation of IDS Rep vs Client Rep

IDS Reps (Blue/Cyan staff badges) must never be conflated with Client Reps (Amber customer badges). Dropdowns must cleanly isolate field inspectors from customer contacts.

RULE 5 Location-Based Currency Assignment

US Plants & Clients evaluate in **USD** (\$X/hr USD). Canadian Plants & Clients evaluate in **CAD** (\$X/hr CAD). Currency tags must always be explicitly rendered.

05 KEY SCREEN ARCHITECTURES

A. THE 1-STOP OPERATIONS HUB (DESKTOP WEB)

ADMIN COMMAND

Unifies Client Onboarding, User Provisioning, Project Setup, and Rep Dispatch into a single control surface:

- **Universal Header:** Instant 1-click triggers (+ Onboard Client & Plant, + Create New User, Assign Rep Dispatch, Export PDF).
- **4-Step Visual Lifecycle Banner:** Step 1 (Client/Plant) → Step 2 (User Logins) → Step 3 (Dispatch & Rates) → Step 4 (Shift Reports & Portal).
- **Master Operations Matrix:** Connected table linking Client, Plant Location + Currency, Scope/Part, PO Cap, Assigned Rep, Billing Rate, and Pay Rate.

B. MOBILE FIELD INSPECTOR APP (375PX VIEWPORT)

PLANT FLOOR

Optimized for high-speed single-thumb execution on mobile screens and Android native APK:

- **Clock-In Hero:** Large green shift launcher button with live running hours counter.
- **4-Step Incident Flow:** Step 1 Details → Step 2 Defect Quantities → Step 3 Action + Camera Canvas → Step 4 Client Routing.
- **Offline Sync Indicator:** Persistent badge showing offline queue status with auto-sync retry.

C. CLIENT EXECUTIVE PORTAL

CUSTOMER VIEW

Dedicated portal for automotive Tier-1 supplier quality managers:

- **Containment Feed:** Published inspection reports with real-time pass/fail counters.
- **Lot Traceability:** Status indicators for certified clean lots and active containment tags.
- **Overtime Approval Drawer:** 1-Click customer sign-off for inspector overtime requests.

06 ENGINEERING COLLABORATION & QUALITY STANDARDS

Design prototypes are seamlessly translated into production components using a centralized design system:

COMPONENT ARCHITECTURE

STANDARDS

- **Atomic Components:** Shared buttons, badges, modals, and tables in dedicated component directories.
- **High-Contrast Enforcement:** Contrast ratio $\geq 7:1$ across all themes and interactive states.
- **Responsive Breakpoints:** Seamless layout scaling from mobile viewports (375px) to dual-monitor desktop (1920px).

QUALITY ASSURANCE

ENTERPRISE

- **Visual Regression Guard:** Automated theme contrast and modal accessibility validation.
- **Zero-Clipping Standard:** Multiline text safety across all on-screen tables and exported reports.
- **Production Verification:** Continuous testing and instantaneous live deployment.